

# Project Title

## SupplyChain Nexus: An Intelligent Supply Chain Risk Analytics and Decision Support Platform

## Problem Statement

Modern supply chains require balancing warehouse capacities, transportation costs, delivery times, and regional demand while minimizing operational risks. Existing analytical tools primarily provide descriptive insights through dashboards, requiring managers to manually interpret data and make decisions. As supply networks grow in complexity, identifying bottlenecks and determining the most efficient distribution strategy becomes increasingly challenging.

## Abstract

SupplyChain Nexus is an intelligent Supply Chain Risk Analytics and Decision Support Platform that assists organizations in analyzing warehouse utilization, identifying logistics bottlenecks, and optimizing distribution strategies using operational supply chain data. The platform integrates warehouse capacities, regional demand, transportation costs, travel times, distance matrices, and geographical information to provide a comprehensive understanding of the supply chain network.

The system computes warehouse utilization, evaluates operational risk based on capacity and demand distribution, and identifies inefficient logistics patterns. A multi-objective optimization engine recommends improved warehouse allocations by minimizing transportation cost and delivery time while balancing warehouse utilization across the network.

To support strategic planning, the platform incorporates a Scenario Analysis Engine that allows decision-makers to evaluate the effects of warehouse outages, capacity reductions, and demand fluctuations. For each simulated scenario, the system recalculates warehouse assignments, logistics costs, delivery times, and utilization levels, enabling organizations to proactively assess risks and compare alternative distribution strategies.

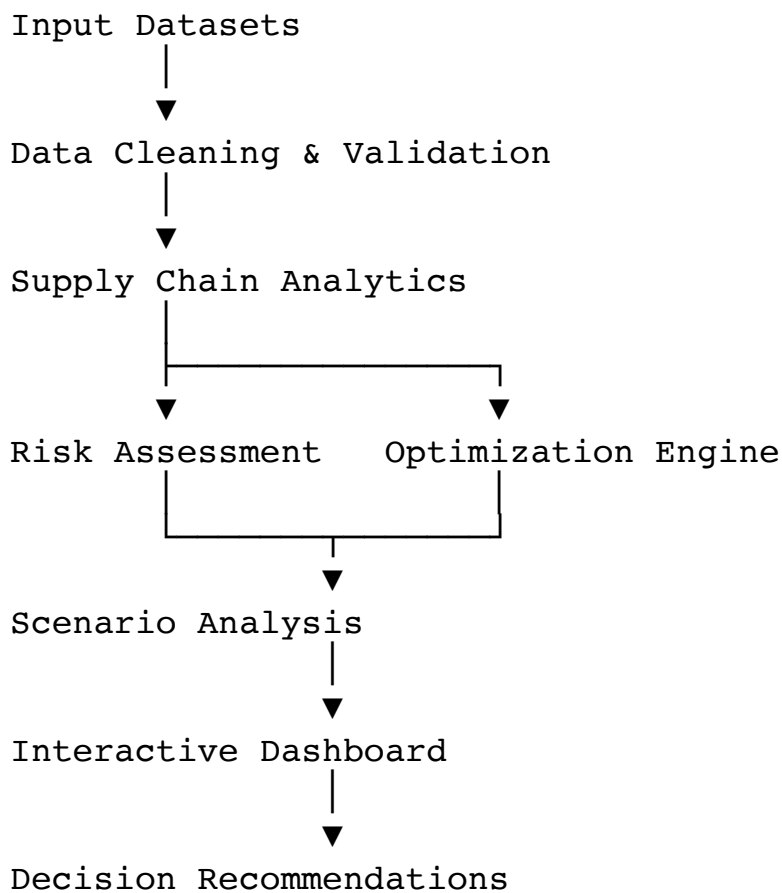
The platform presents these insights through an interactive geospatial dashboard featuring warehouse locations, optimized transportation routes, utilization heatmaps, operational alerts, and AI-generated recommendations. By combining analytics, optimization, simulation, and visualization into a unified platform, SupplyChain Nexus enables organizations to make faster, data-driven decisions that improve operational efficiency, reduce logistics costs, and strengthen supply chain resilience.

## Objectives

- Analyze warehouse utilization across the network.
- Identify operational bottlenecks and capacity constraints.

- Optimize warehouse allocation based on multiple objectives.
- Minimize transportation cost and delivery time.
- Improve warehouse resource utilization.
- Simulate operational scenarios for proactive planning.
- Assist decision-makers with intelligent recommendations.

## Proposed Workflow



## Modules

### Module 1: Data Integration

Integrates all provided datasets:

- Demand
- Warehouse Capacity
- Transportation Cost

- Distance Matrix
- Travel Time
- Geographic Coordinates

## **Module 2: Warehouse Utilization Analysis**

Calculates

- Current utilization
- Available capacity
- Capacity distribution
- Warehouse occupancy

Output

- Utilization charts
- Capacity comparison
- Warehouse status

## **Module 3: Supply Chain Risk Assessment**

Computes warehouse risk based on

- Capacity utilization
- Distance from demand centers
- Transportation cost
- Delivery time

Each warehouse receives a

- Low Risk
- Medium Risk
- High Risk

classification.

## **Module 4: Multi-Objective Optimization Engine**

Optimizes warehouse allocation considering

- Transportation Cost
- Delivery Time
- Warehouse Capacity

The system recommends

- Best warehouse
- Best allocation
- Balanced resource utilization

## **Module 5: Scenario Analysis Engine**

Allows users to simulate

### **Scenario 1**

Increase regional demand

System recalculates

- warehouse utilization
- logistics cost
- delivery time

### **Scenario 2**

Warehouse capacity reduction

System identifies

- overloaded warehouses
- alternate warehouse allocation

### **Scenario 3**

Warehouse outage

System redistributes demand to available warehouses and compares the updated network with the original configuration.

## Module 6: Geospatial Visualization

Interactive map displaying

- Warehouse locations
- Demand regions
- Transportation routes
- Optimized routes
- Risk heatmaps

## Module 7: Recommendation Engine

Automatically suggests

- Best warehouse allocation
- Alternate warehouse
- Capacity balancing
- Logistics improvements

Example

Recommendation

Warehouse W3 is operating at 95% utilization.

Redistribute Region R7 to Warehouse W5.

Expected Benefits

Transportation Cost ↓ 11%

Warehouse Utilization ↓ 18%

Delivery Time ↓ 9%

## Module 8: Comparative Analytics

Displays

**Current Network**

- Total Cost
- Average Delivery Time
- Warehouse Utilization

## **Optimized Network**

- Total Cost
- Average Delivery Time
- Warehouse Utilization

Shows measurable improvements.

# **Technologies**

## **Frontend**

- React
- Tailwind CSS
- Leaflet
- Plotly

## **Backend**

- FastAPI

## **Optimization**

- Google OR-Tools
- PuLP (Linear Programming)

## **Data Processing**

- Pandas
- NumPy

## **Database**

- SQLite

# Expected Outcomes

- Reduced transportation cost
- Improved warehouse utilization
- Reduced delivery time
- Better warehouse allocation
- Faster operational decision-making
- Enhanced supply chain resilience

# Novelty

Unlike traditional dashboards that only visualize operational data, SupplyChain Nexus combines **risk assessment, multi-objective optimization, scenario analysis, and decision recommendations** into a single platform. The system not only identifies inefficiencies but also recommends optimized warehouse allocations and allows decision-makers to evaluate alternative logistics strategies before implementation.

# Why this project is feasible

Every major component is directly supported by your datasets:

| Feature                      | Dataset Used                               |
|------------------------------|--------------------------------------------|
| Warehouse Utilization        | Capacity + Demand                          |
| Transportation Cost Analysis | Cost                                       |
| Delivery Time Analysis       | Time                                       |
| Route Analysis               | Distance + GeoLocations                    |
| Warehouse Allocation         | Capacity + Demand + Cost                   |
| Multi-Objective Optimization | Capacity + Cost + Time + Distance          |
| Risk Assessment              | Capacity + Demand + Time + Cost            |
| Scenario Analysis            | Modified values from the provided datasets |
| Interactive Map              | GeoLocations                               |

This ensures the project remains fully aligned with the hackathon constraints while demonstrating a complete engineering solution rather than just a visualization dashboard.